EXERCISE (O-1)

SINGLE CORRECT TYPE QUESTIONS

1. Eccentricity of the hyperbola conjugate to the hyperbola $\frac{x^2}{4}-\frac{y^2}{12}=1$ is

(A) $\frac{2}{\sqrt{3}}$ (B) 2

(C) $\sqrt{3}$ (D) $\frac{4}{3}$

MHB135

2. The foci of a hyperbola coincide with the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{9}=1$. Then the equation of the hyperbola with eccentricity 2 is

(A) $\frac{x^2}{12}-\frac{y^2}{4}=1$ (B) $\frac{x^2}{4}-\frac{y^2}{12}=1$

(C) $3x^2 - y^2 + 12 = 0$ (D) $9x^2 - 25y^2 - 225 = 0$

MHB136

3. If the eccentricity of the hyperbola $x^2 - y^2 \sec^2\alpha = 5$ is $\sqrt{3}$ times the eccentricity of the ellipse $x^2\sec^2\alpha + y^2 = 25$, then a value of α is :

(A) $\pi/6$ (B) $\pi/4$

(C) $\pi/3$ (D) $\pi/2$

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MHB137

4. The foci of the ellipse $\frac{x^2}{16}+\frac{y^2}{b^2}=1$ and the hyperbola $\frac{x^2}{144}-\frac{y^2}{81}=\frac{1}{25}$ coincide. Then the value of b^2 is-

(A) 5 (B) 7 (C) 9 (D) 4

MHB138

5. The graph of the equation $x + y = x^3 + y^3$ is the union of -

(A) line and an ellipse (B) line and a parabola

(C) line and hyperbola (D) line and a point

MHB139

6. The focal length of the hyperbola $x^2 - 3y^2 - 4x - 6y - 11 = 0$, is -

(A) 4 (B) 6 (C) 8 (D) 10

MHB140

7. The equation $\frac{x^2}{29-p}+\frac{y^2}{4-p}=1$ $(p \neq 4, 29)$ represents -

(A) an ellipse if p is any constant greater than 4

(B) a hyperbola if p is any constant between 4 and 29.

(C) a rectangular hyperbola if p is any constant greater than 29.

(D) no real curve is p is less than 29.

MHB141

For More Join: @IITJEE_Advanced

8. If $\frac{x^2}{\cos^2\alpha}-\frac{y^2}{\sin^2\alpha}=1$ represents family of hyperbolas where 'α' varies then -

(A) distance between the foci is constant

(B) distance between the two directrices is constant

(C) distance between the vertices is constant

(D) distances between focus and the corresponding directrix is constant

MHB142

9. The locus of the point of intersection of the lines $\sqrt{3}x - y - 4\sqrt{3}t = 0$ & $\sqrt{3}tx + ty - 4\sqrt{3} = 0$ (where t is a parameter) is a hyperbola whose eccentricity is

(A) $\sqrt{3}$ (B) 2 (C) $\frac{2}{\sqrt{3}}$ (D) $\frac{4}{3}$

MHB143

10. Latus rectum of the conic satisfying the differential equation, $x\,dy + y\,dx = 0$ and passing through the point (2, 8) is :

(A) $4\sqrt{2}$ (B) 8 (C) $8\sqrt{2}$ (D) 16

MHB144

11. The magnitude of the gradient of the tangent at an extremity of latera recta of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is equal to (where e is the eccentricity of the hyperbola)

(A) be (B) e (C) ab (D) ae

MHB145

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12. The number of possible tangents which can be drawn to the curve $4x^2 - 9y^2 = 36$, which are perpendicular to the straight line $5x + 2y - 10 = 0$ is :

(A) zero (B) 1 (C) 2 (D) 4

MHB146

13. Locus of the point of intersection of the tangents at the points with eccentric angles ϕ and $\frac{\pi}{2}-\phi$ on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ is :

(A) $x = a$ (B) $y = b$ (C) $x = ab$ (D) $y = ab$

MHB147

14. Locus of the feet of the perpendiculars drawn from either foci on a variable tangent to the hyperbola $16y^2 - 9x^2 = 1$ is

(A) $x^2 + y^2 = 9$ (B) $x^2 + y^2 = 1/9$

(C) $x^2 + y^2 = 7/144$ (D) $x^2 + y^2 = 1/16$

MHB148

15. A tangent to the ellipse $\frac{x^2}{9}+\frac{y^2}{4}=1$ with centre C meets its director circle at P and Q. Then the product of the slopes of CP and CQ, is -

(A) $\frac{9}{4}$ (B) $\frac{-4}{9}$ (C) $\frac{2}{9}$ (D) $-\frac{1}{4}$

MHB149

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16. The asymptote of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ form with any tangent to the hyperbola a triangle whose area is $a^2\tan\lambda$ in magnitude then its eccentricity is :

(A) $\sec\lambda$ (B) $\operatorname{cosec}\lambda$ (C) $\sec^2\lambda$ (D) $\operatorname{cosec}^2\lambda$

MHB150

17. In which of the following cases maximum number of normals can be drawn from a point P lying in the same plane

(A) circle (B) parabola (C) ellipse (D) hyperbola

MHB151

18. PQ is a double ordinate of the ellipse $x^2 + 9y^2 = 9$, the normal at P meets the diameter through Q at R, then the locus of the mid point of PR is

(A) a circle (B) a parabola (C) an ellipse (D) a hyperbola

MHB152

19. With one focus of the hyperbola $\frac{x^2}{9}-\frac{y^2}{16}=1$ as the centre, a circle is drawn which is tangent to the hyperbola with no part of the circle being outside the hyperbola. The radius of the circle is

(A) less than 2 (B) 2 (C) $\frac{11}{3}$ (D) none

MHB153

20. If the normal to the rectangular hyperbola $xy = c^2$ at the point 't' meets the curve again at 't_1' then t^3t_1 has the value equal to

(A) 1 (B) – 1 (C) 0 (D) none

MHB154

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21. Number of common tangent with finite slope to the curves $xy = c^2$ & $y^2 = 4ax$ is :

(A) 0 (B) 1 (C) 2 (D) 4

MHB155

22. Locus of the middle points of the parallel chords with gradient m of the rectangular hyperbola $xy = c^2$ is

(A) $y + mx = 0$ (B) $y - mx = 0$ (C) $my - x = 0$ (D) $my + x = 0$

MHB156

23. Let F_1, F_2 are the foci of the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ and F_3, F_4 are the foci of its conjugate hyperbola. If e_H and e_C are their eccentricities respectively then the statement which holds true is

(A) Their equations of the asymptotes are different.

(B) $e_H > e_C$

(C) Area of the quadrilateral formed by their foci is 50 sq. units.

(D) Their auxiliary circles will have the same equation.

MHB157

24. AB is a double ordinate of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ such that ΔAOB (where 'O' is the origin) is an equilateral triangle, then the eccentricity e of the hyperbola satisfies

(A) $e > \sqrt{3}$ (B) $1 < e < \frac{2}{\sqrt{3}}$ (C) $e = \frac{2}{\sqrt{3}}$ (D) $e > \frac{2}{\sqrt{3}}$

MHB158

For More Join: @IITJEE_Advanced

25. P is a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, N is the foot of the perpendicular from P on the transverse axis. The tangent to the hyperbola at P meets the transverse axis at T. If O is the centre of the hyperbola, then $OT.ON$ is equal to :

(A) e^2 (B) a^2 (C) b^2 (D) b^2/a^2

MHB159

26. Let the major axis of a standard ellipse equals the transverse axis of a standard hyperbola and their director circles have radius equal to $2R$ and R respectively. If e_1 and e_2 are the eccentricities of the ellipse and hyperbola then the correct relation is

(A) $4e_1^2 - e_2^2 = 6$ (B) $e_1^2 - 4e_2^2 = 2$ (C) $4e_2^2 - e_1^2 = 6$ (D) $2e_1^2 - e_2^2 = 4$

MHB160

27. The tangent to the hyperbola $xy = c^2$ at the point P intersects the x-axis at T and the y-axis at T'. The normal to the hyperbola at P intersects the x-axis at N and the y-axis at N'. The areas of the triangles PNT and $PN'T'$ are Δ and Δ' respectively, then $\frac{1}{\Delta}+\frac{1}{\Delta'}$ is

(A) equal to 1 (B) depends on t (C) depends on c (D) equal to 2

MHB161

28. The chord PQ of the rectangular hyperbola $xy = a^2$ meets the axis of x at A ; C is the mid point of PQ & 'O' is the origin. Then the ΔACO is:

(A) equilateral (B) isosceles (C) right angled (D) right isosceles.

MHB162

29. The locus of the foot of the perpendicular from the centre of the hyperbola $xy = c^2$ on a variable tangent is :

(A) $(x^2 - y^2)^2 = 4c^2 xy$ (B) $(x^2 + y^2)^2 = 2c^2 xy$

(C) $(x^2 + y^2) = 4c^2 xy$ (D) $(x^2 + y^2)^2 = 4c^2 xy$

MHB163

30. The equation to the chord joining two points (x_1, y_1) and (x_2, y_2) on the rectangular hyperbola $xy = c^2$ is :

(A) $\frac{x}{x_1+x_2}+\frac{y}{y_1+y_2}=1$ (B) $\frac{x}{x_1-x_2}+\frac{y}{y_1-y_2}=1$

(C) $\frac{x}{y_1+y_2}+\frac{y}{x_1+x_2}=1$ (D) $\frac{x}{y_1-y_2}+\frac{y}{x_1-x_2}=1$

MHB164

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EXERCISE (O-2)

MULTIPLE CORRECT TYPE QUESTIONS

1. Let p and q be non-zero real numbers. Then the equation $(px^2 + qy^2 + r)(4x^2 + 4y^2 - 8x - 4) = 0$ represents

(A) two straight lines and a circle, when $r = 0$ and p, q are of the opposite sign.

(B) two circles, when $p = q$ and r is of sign opposite to that of p.

(C) a hyperbola and a circle, when p and q are of opposite sign and $r \neq 0$.

(D) a circle and an ellipse, when p and q are unequal but of same sign and r is of sign opposite to that of p.

MHB165

2. If θ is eliminated from the equations $a \sec\theta - x \tan\theta = y$ and $b \sec\theta + y \tan\theta = x$ (a and b are constant), then the eliminant denotes the equation of

(A) the director circle of the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$

(B) auxiliary circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(C) Director circle of the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

(D) Director circle of the circle $x^2 + y^2 = \frac{a^2 + b^2}{2}$.

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MHB166

3. The tangent to the hyperbola, $x^2 - 3y^2 = 3$ at the point $\left(\sqrt{3}, 0\right)$ when associated with two asymptotes constitutes -

(A) isosceles triangle which is not equilateral

(B) an equilateral triangle

(C) a triangles whose area is $\sqrt{3}$ sq. units

(D) a right isosceles triangle.

MHB167

4. If latus rectum of a hyperbola subtends a right angle at other focus of hyperbola, then eccentricity is equal to-

(A) $1 - \sqrt{2}$ (B) $\tan\frac{\pi}{8}$ (C) $\cot\frac{\pi}{8}$ (D) $\left(\frac{1}{\sqrt{2}-1}\right)$

MHB168

5. If the circle $x^2 + y^2 = a^2$ intersects the hyperbola $xy = c^2$ in four points $P(x_1, y_1)$, $Q(x_2, y_2)$, $R(x_3, y_3)$, $S(x_4, y_4)$, then -

(A) $x_1 + x_2 + x_3 + x_4 = 0$ (B) $y_1 + y_2 + y_3 + y_4 = 0$

(C) $x_1x_2x_3x_4 = c^4$ (D) $y_1y_2y_3y_4 = c^4$

MHB169

For More Join: @IITJEE_Advanced

6. Which of the following equations in parametric form can represent a hyperbolic profile, where 't' is a parameter.

(A) $x = \frac{a}{2}\left(t + \frac{1}{t}\right)$ & $y = \frac{b}{2}\left(t - \frac{1}{t}\right)$

(B) $\frac{tx}{a} - \frac{y}{b} + t = 0$ & $\frac{x}{a} + \frac{ty}{b} - 1 = 0$

(C) $x = e^t + e^{-t}$ & $y = e^t - e^{-t}$

(D) $x^2 - 6 = 2\cos t$ & $y^2 + 2 = 4\cos^2\frac{t}{2}$

MHB170

7. Let $A(-1,0)$ and $B(2,0)$ be two points on the x-axis. A point M is moving in xy-plane (other than x-axis) in such a way that $\angle MBA = 2\angle MAB$, then the point M moves along a conic whose

(A) coordinate of vertices are (± 3, 0).

(B) length of latus-rectum equals 6.

(C) eccentricity equals 2.

(D) equation of directrices are $x = \pm\frac{1}{2}$.

MHB171

8. Hyperbola $\frac{x^2}{a^2} - \frac{y^2}{3} = 1$ of eccentricity e is confocal with the ellipse $\frac{x^2}{8} + \frac{y^2}{4} = 1$. Let A,B,C & D are points of intersection of hyperbola & ellipse, then-

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(A) $e = \frac{5}{2}$

(B) $e = 2$

(C) A,B,C,D are concyclic points

(D) Number of common tangents of hyperbola & ellipse is 2

MHB172

9. If the ellipse $4x^2 + 9y^2 + 12x + 12y + 5 = 0$ is confocal with a hyperbola having same principal axes, then -

(A) angle between normals at their each point of intersection is 90°.

(B) centre of the ellipse is $\left(-\frac{3}{2}, -\frac{2}{3}\right)$

(C) distance between foci of the hyperbola is $\frac{2\sqrt{10}}{3}$

(D) ellipse and hyperbola has same length of latus rectum

MHB173

10. If the eccentricity of the ellipse $\frac{x^2}{(\log a)^2} + \frac{y^2}{(\log b)^2} = 1$ $(a > b > 0,\ a,\ b \neq 1)$ is $\frac{1}{\sqrt{2}}$ and 'e' be the eccentricity of the hyperbola $\frac{x^2}{(\log_b a)^2} - y^2 = 1$, then e^2 is greater than (where $\log x = \ell nx$)-

(A) $\frac{3}{2}$ (B) $\frac{1}{2}$ (C) $\frac{2}{3}$ (D) $\frac{5}{4}$

MHB174

For More Join: @IITJEE_Advanced

11. Let the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be reciprocal to that of the ellipse $x^2 + 4y^2 = 4$. If the hyperbola passes through a focus of the ellipse, then -

(A) the equation of the hyperbola is $\frac{x^2}{3}-\frac{y^2}{2}=1$

(B) a focus of the hyperbola is (2, 0)

(C) the eccentricity of the hyperbola is $\sqrt{\frac{5}{3}}$

(D) the equation of the hyperbola is $x^2 - 3y^2 = 3$

MHB175

12. Tangents are drawn to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$, parallel to the straight line $2x - y = 1$. The points of contact of the tangents on the hyperbola are

(A) $\left(\frac{9}{2\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ (B) $\left(-\frac{9}{2\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$

(C) $\left(3\sqrt{3},-2\sqrt{2}\right)$ (D) $\left(-3\sqrt{3},2\sqrt{2}\right)$

MHB176

13. An ellipse intersects the hyperbola $2x^2 - 2y^2 = 1$ orthogonally. The eccentricity of the ellipse is reciprocal of that of the hyperbola. If the axes of the ellipse are along the coordinate axes, then

(A) Equation of ellipse is $x^2 + 2y^2 = 2$ (B) The foci of ellipse are $(\pm 1, 0)$

(C) Equation of ellipse is $x^2 + 2y^2 = 4$ (D) The foci of ellipse are $(\pm \sqrt{2}, 0)$

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MHB177

14. If the normal at point P to the rectangular hyperbola $x^2 - y^2 = 4$ meets the transverse and conjugate axes at A and B respectively and C is the centre of the hyperbola, then -

(A) $PA = PC$ (B) $PA = PB$

(C) $PB = PC$ (D) $AB = 2PC$

MHB178

15. Let an incident ray $L_1 = 0$ gets reflected at point $A(-2, 3)$ on hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ and passes through focus $S(2, 0)$, then -

(A) equation of incident ray is $x + 2 = 0$

(B) equation of reflected ray is $3x + 4y = 6$

(C) eccentricity, $e = 2$

(D) length of latus rectum = 6

MHB179

16. All the points on the asymptotes of a certain hyperbola satisfy the equation of $x^2 - 4y^2 = 0$ then the possible value(s) of eccentricity of the hyperbola is/are -

(A) $\frac{\sqrt{5}}{2}$ (B) $\frac{\sqrt{5}}{3}$ (C) $\sqrt{5}$ (D) $\frac{3}{2}$

MHB180

For More Join: @IITJEE_Advanced

17. If the equation of a hyperbola is given as $\left|\sqrt{(x-3)^2+(y-3)^2}-\sqrt{(x-1)^2+(y-1)^2}\right|=2$, then-

(A) The given hyperbola is a rectangular hyperbola.
(B) Distance between both directrices is $\sqrt{2}$
(C) Centre of hyperbola lies on line $y = x$
(D) Asymptotes are inclined at an angle of 45° with each other.

MHB181

18. Let the variable line $y = kx + h$ is tangent to the hyperbola $\frac{x^2}{4}-\frac{y^2}{9}=1$. If locus of $P(h,k)$ is a conic then which of the following is true?
(A) focus of conic lies on y-axis.
(B) eccentricity of conic is greater than the eccentricity of given hyperbola
(C) centre of conic is at origin
(D) conic is an ellipse

MHB182

19. If the ellipse $x^2 + 2y^2 = 4$ touches the hyperbola $xy = c$ at points P and Q, then -
(A) c is a rational number
(B) O,P and Q are collinear points (where 'O' is origin)
(C) length of PQ is $2\sqrt{3}$
(D) length of PQ is $2\sqrt{2}$

MHB183

20. If (3,4) and (8,6) are foci of a conic passing through origin, then eccentricity of conic is-
(A) $\frac{\sqrt{29}}{16}$ (B) $\frac{\sqrt{29}}{15}$ (C) $\frac{\sqrt{29}}{12}$ (D) $\frac{\sqrt{29}}{5}$

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MHB184

COMPREHENSION TYPE QUESTIONS

Paragraph for question nos. 21 to 23

From a point 'P' three normals are drawn to the parabola $y^2 = 4x$ such that two of them make angles with the abscissa axis, the product of whose tangents is 2. Suppose the locus of the point 'P' is a part of a conic 'C'. Now a circle $S = 0$ is described on the chord of the conic 'C' as diameter passing through the point (1, 0) and with gradient unity. Suppose (a, b) are the coordinates of the centre of this circle. If L_1 and L_2 are the two asymptotes of the hyperbola with length of its transverse axis $2a$ and conjugate axis $2b$ (principal axes of the hyperbola along the coordinate axes) then answer the following questions.

21. Locus of P is a
(A) circle (B) parabola (C) ellipse (D) hyperbola

MHB185

22. Radius of the circle $S = 0$ is
(A) 4 (B) 5 (C) $\sqrt{17}$ (D) $\sqrt{23}$

MHB186

23. The angle $\alpha \in (0, \pi/2)$ between the two asymptotes of the hyperbola lies in the interval
(A) (0, 15°) (B) (30°, 45°) (C) (45°, 60°) (D) (60°, 75°)

MHB187

For More Join: @IITJEE_Advanced

Paragraph for question nos. 24 to 26

Consider the conic $C: \frac{x^2}{16}+\frac{y^2}{12}=1$

24. Equation of circle touching C at one extremity of latus-rectum and passing through centre of C is/are-

(A) $8x^2 + 8y^2 - 19x - 22y = 0$

(B) $8x^2 + 8y^2 - 19x + 22y = 0$

(C) $8x^2 + 8y^2 + 19x - 22y = 0$

(D) $8x^2 + 8y^2 + 19x + 22y = 0$

MHB188

25. The equation of parabolas with same latus-rectum as conic C, is/are-

(A) $y^2 - 6x + 3 = 0$

(B) $y^2 + 6x - 21 = 0$

(C) $y^2 - 6x - 21 = 0$

(D) $y^2 + 6x + 3 = 0$

MHB189

26. If a hyperbola passes through extremities of minor axis of conic C and its transverse and conjugate axis coincides with the minor and major axis of conic C respectively, and the product of eccentricity of hyperbola and conic C is 1, then -

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(A) equation of hyperbola is $\frac{x^2}{36}-\frac{y^2}{12}=-1$.

(B) equation of hyperbola is $\frac{x^2}{9}-\frac{y^2}{3}=-1$.

(C) focus of hyperbola is $(0,2\sqrt{3})$

(D) focus of hyperbola is $(0,-4\sqrt{3})$

MHB190

Paragraph for question nos. 27 to 28

The graph of the conic $x^2 - (y - 1)^2 = 1$ has one tangent line with positive slope that passes through the origin. the point of tangency being (a, b). Then

27. The value of $\sin^{-1}\left(\frac{a}{b}\right)$ is $\lambda\pi$, then λ is

MHB191

28. Length of the latus rectum of the conic is

MHB192

For More Join: @IITJEE_Advanced

MATCHING LIST TYPE QUESTION

29.

	List-I		**List-II**
(P)	Value of c for which $3x^2 - 5xy - 2y^2 + 5x + 11y + c = 0$ are the asymptotes of the hyperbola $3x^2 - 5xy - 2y^2 + 5x + 11y - 8 = 0$	(1)	3
(Q)	If locus of a point, whose chord of contact with respect to the circle $x^2 + y^2 = 4$ is a tangent to the hyperbola $xy = 1$ is $xy = c^2$, then value of c^2 is	(2)	–4
(R)	If equation of a hyperbola whose conjugate axis is 5 and distance between its foci is 13, is $ax^2 - by^2 = c$ where a and b are coprime natural numbers, then value of $\frac{ab}{c}$ is	(3)	–12
		(4)	4
(S)	If the vertex of a hyperbola bisects the distance between its centre and the corresponding focus, then ratio of square of its conjugate axis to the square of its transverse axis is	(5)	–6

Which of the following is correct option ?

(A) P $\rightarrow$ 2; Q $\rightarrow$ 1; R $\rightarrow$ 4; S $\rightarrow$ 1 (B) P $\rightarrow$ 3; Q $\rightarrow$ 4; R $\rightarrow$ 1; S $\rightarrow$ 5

(C) P $\rightarrow$ 3; Q $\rightarrow$ 4; R $\rightarrow$ 4; S $\rightarrow$ 1 (D) P $\rightarrow$ 1; Q $\rightarrow$ 4; R $\rightarrow$ 5; S $\rightarrow$ 3

MHB193

MATRIX MATCH TYPE QUESTION

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30.

	Column-I		**Column-II**
(A)	Number of positive integral values of b for which tangent parallel to line $y = x + 1$ can be drawn to hyperbola $\frac{x^2}{5} - \frac{y^2}{b^2} = 1$ is	(P)	16
(B)	The equation of the hyperbola with vertices (3, 0) and (–3, 0) and semi-latusrectum 4, is given by is $4x^2 - 3y^2 = 4k$, then k =	(Q)	2
(C)	The product of the lengths of the perpendiculars from the two focii on any tangent to the hyperbola $\frac{x^2}{25} - \frac{y^2}{3} = 1$ is $\sqrt{k}$, then k is	(R)	4
(D)	An equation of a tangent to the hyperbola, $16x^2 - 25y^2 - 96x + 100y - 356 = 0$ which makes an angle $\frac{\pi}{4}$ with the transverse axis is $y = x + \lambda$, $(\lambda > 0)$, then 2λ is	(S)	9

MHB194

For More Join: @IITJEE_Advanced

EXERCISE (S)

1. The hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ $(a,b>0)$ passes through the point of intersection of the lines, $7x + 13y - 87 = 0$ & $5x - 8y + 7 = 0$ and the latus rectum is $\frac{32\sqrt{2}}{5}$, then value of b is equal to

MHB016

2. The line $2x + y = 1$ is tangent to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If this line passes through the point of intersection of the nearest directrix and the x-axis, then the eccentricity of the hyperbola is

MHB017

3. Tangents to the hyperbola $x^2 - 9y^2 = 9$ that are drawn from (3, 2). Find the area of the triangle that these tangents form with their chord of contact.

MHB018

4. For the hyperbola $\frac{x^2}{100}-\frac{y^2}{25}=1$, the value of $\sqrt{SA\cdot S'A}$ is equal to, (where S & S' are the foci & A is the vertex)

MHB019

5. If tangents drawn from the point $(a, 2)$ to the hyperbola $\frac{x^2}{16}-\frac{y^2}{9}=1$ are perpendicular, then the value of a^2 is

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MHB020

6. From the centre C of the hyperbola $x^2 - y^2 = 9$, CM is drawn perpendicular to the tangent at any point of the curve, meeting the tangent at M and the curve at N. Find the value of the product $(CM)(CN)$.

MHB057

7. If the lines $x + y + 1 = 0$ and $2x - y + 2 = 0$ are the asymptotes of a hyperbola. If the line $x - 2 = 0$ touches the hyperbola then the equation of the hyperbola is $4(x + y + 1)(2x - y + 2) = \lambda$. Find the value of λ.

MHB058

8. Tangent and normal are drawn at the upper end (x_1, y_1) of the latus rectum P with $x_1 > 0$ and $y_1 > 0$, of the hyperbola $\frac{x^2}{4}-\frac{y^2}{12}=1$, intersecting the transverse axis at T and G respectively. Find the area of the triangle PTG.

MHB059

9. If $7x^2 + pxy + qy^2 + rx - sy + t = 0$ is the eqaution of the hyperbola whose one focus is (–1, 1), eccentricity = 3 and the equation of the corresponding directrix is $x - y + 3 = 0$, then the value of 't' is:

MHB073

10. Tangents are drawn to the hyperbola $4x^2 - y^2 = 36$ at the point P and Q. If these tangents intersect at the point $T(0, 3)$ then the area (in sq. units) of ΔPTQ is -

MHB074

For More Join: @IITJEE_Advanced

EXERCISE - JEE (Main) PYQ

1. Let $0 < \theta < \frac{\pi}{2}$. If the eccentricity of the hyperbola $\frac{x^2}{\cos^2\theta} - \frac{y^2}{\sin^2\theta} = 1$ is greater than 2, then the length of its latus rectum lies in the interval : **[JEE (Main) 2019]**

(1) (2, 3] (2) $(3, \infty)$ (3) (3/2, 2] (4) (1, 3/2]

MHB029

2. The equation of a tangent to the hyperbola $4x^2 - 5y^2 = 20$ parallel to the line $x - y = 2$ is: **[JEE (Main) 2019]**

(1) $x - y + 9 = 0$ (2) $x - y + 7 = 0$ (3) $x - y + 1 = 0$ (4) $x - y - 3 = 0$

MHB030

3. Let $S = \left\{(x, y) \in R^2 : \frac{y^2}{1+r} - \frac{x^2}{1-r} = 1\right\}$, where $r \neq \pm 1$. Then S represents: **[JEE (Main) 2019]**

(1) A hyperbola whose eccentricity is $\frac{2}{\sqrt{r+1}}$, where $0 < r < 1$.

(2) An ellipse whose eccentricity is $\frac{1}{\sqrt{r+1}}$, where $r > 1$

(3) A hyperbola whose eccentricity is $\frac{2}{\sqrt{1-r}}$, when $0 < r < 1$.

(4) An ellipse whose eccentricity is $\sqrt{\frac{2}{r+1}}$, when $r > 1$

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MHB031

4. If a hyperbola has length of its conjugate axis equal to 5 and the distance between its foci is 13, then the eccentricity of the hyperbola is :- **[JEE (Main) 2019]**

(1) 2 (2) $\frac{13}{6}$ (3) $\frac{13}{8}$ (4) $\frac{13}{12}$

MHB032

5. If the line $y = mx + 7\sqrt{3}$ is normal to the hyperbola $\frac{x^2}{24} - \frac{y^2}{18} = 1$, then a value of m is **[JEE (Main) 2019]**

(1) $\frac{\sqrt{5}}{2}$ (2) $\frac{3}{\sqrt{5}}$ (3) $\frac{2}{\sqrt{5}}$ (4) $\frac{\sqrt{15}}{2}$

MHB033

6. If a hyperbola passes through the point $P(10,16)$ and it has vertices at $(\pm 6, 0)$, then the equation of the normal to it at P is **[JEE (Main) 2020]**

(1) $x + 2y = 42$ (2) $3x + 4y = 94$

(3) $2x + 5y = 100$ (4) $x + 3y = 58$

MHB034

For More Join: @IITJEE_Advanced

7. A line parallel to the straight line $2x-y=0$ is tangent to the hyperbola $\frac{x^2}{4}-\frac{y^2}{2}=1$ at the point (x_1, y_1). Then $x_1^2+5y_1^2$ is equal to : **[JEE (Main) 2020]**

(1) 5 (2) 6 (3) 8 (4) 10

MHB035

8. A hyperbola having the transverse axis of length $\sqrt{2}$ has the same foci as that of the ellipse $3x^2+4y^2=12$, then this hyperbola does not pass through which of the following points? **[JEE (Main) 2020]**

(1) $\left(1,-\frac{1}{\sqrt{2}}\right)$ (2) $\left(\sqrt{\frac{3}{2}},\frac{1}{\sqrt{2}}\right)$ (3) $\left(\frac{1}{\sqrt{2}},0\right)$ (4) $\left(-\sqrt{\frac{3}{2}},1\right)$

MHB036

9. If the line $y=mx+c$ is a common tangent to the hyperbola $\frac{x^2}{100}-\frac{y^2}{64}=1$ and the circle $x^2+y^2=36$, then which one of the following is true? **[JEE (Main) 2020]**

(1) $5m=4$ (2) $4c^2=369$ (3) $c^2=369$ (4) $8m+5=0$

MHB037

10. Let e_1 and e_2 be the eccentricities of the ellipse, $\frac{x^2}{25}+\frac{y^2}{b^2}=1(b<5)$ and the hyperbola, $\frac{x^2}{16}-\frac{y^2}{b^2}=1$ respectively satisfying $e_1e_2 = 1$. If α and β are the distances between the foci of the ellipse and the foci of the hyperbola respectively, then the ordered pair (α, β) is equal to : **[JEE (Main) 2020]**

(1) (8, 10) (2) (8, 12) (3) $\left(\frac{20}{3},12\right)$ (4) $\left(\frac{24}{5},10\right)$

MHB075

11. Let a line $L: 2x+y=k, k>0$ be a tangent to the hyperbola $x^2-y^2=3$. If L is also a tangent to the parabola $y^2 = \alpha x$, then α is equal to: **[JEE (Main) 2021]**

(1) 12 (2) –12 (3) 24 (4) –24

MHB038

12. The locus of the centroid of the triangle formed by any point P on the hyperbola $16x^2-9y^2+32x+36y-164=0$, and its foci is: **[JEE (Main) 2021]**

(1) $16x^2-9y^2+32x+36y-36=0$ (2) $9x^2-16y^2+36x+32y-144=0$

(3) $16x^2-9y^2+32x+36y-144=0$ (4) $9x^2-16y^2+36x+32y-36=0$

MHB039

13. A hyperbola passes through the foci of the ellipse $\frac{x^2}{25}+\frac{y^2}{16}=1$ and its transverse and conjugate axes coincide with major and minor axes of the ellipse, respectively. If the product of their eccentricities in one, then the equation of the hyperbola is : **[JEE (Main) 2021]**

(1) $\frac{x^2}{9}-\frac{y^2}{25}=1$ (2) $\frac{x^2}{9}-\frac{y^2}{16}=1$ (3) $x^2-y^2=9$ (4) $\frac{x^2}{9}-\frac{y^2}{4}=1$

MHB076

For More Join: @IITJEE_Advanced

14. The locus of the point of intersection of the lines $(\sqrt{3})kx+ky-4\sqrt{3}=0$ and $\sqrt{3}x-y-4(\sqrt{3})k=0$ is a conic, whose eccentricity is. **[JEE (Main) 2021]**

MHB077

15. Consider a hyperbola $H: x^2-2y^2=4$. Let the tangent at a point $P\left(4,\sqrt{6}\right)$ meet the x –axis at Q and latus rectum at $R(x_1,y_1), x_1>0$. If F is a focus of H which is nearer to the point P, then the area of ΔQFR is equal to **[JEE (Main) 2021]**

(1) $4\sqrt{6}$ (2) $\sqrt{6}-1$ (3) $\frac{7}{\sqrt{6}}-2$ (4) $4\sqrt{6}-1$

MHB078

16. The normal to the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{9}=1$ at the point $\left(8,3\sqrt{3}\right)$ on it passes through the point: **[JEE (Main) 2022]**

(1) $\left(15,-2\sqrt{3}\right)$ (2) $\left(9,2\sqrt{3}\right)$ (3) $\left(-1,9\sqrt{3}\right)$ (4) $\left(-1,6\sqrt{3}\right)$

MHB040

17. Let the tangent drawn to the parabola $y^2=24x$ at the point (α,β) is perpendicular to the line $2x+2y=5$. Then the normal to the hyperbola $\frac{x^2}{\alpha^2}-\frac{y^2}{\beta^2}=1$ at the point $(\alpha+4,\beta+4)$ does NOT pass through the point : **[JEE (Main) 2022]**

(1) (25, 10) (2) (20, 12) (3) (30, 8) (4) (15, 13)

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MHB041

18. Let the hyperbola $H: \frac{x^2}{a^2}-y^2=1$ and the ellipse $E: 3x^2+4y^2=12$ be such that the length of latus rectum of H is equal to the length of latus rectum of E. If e_H and e_E are the eccentricities of H and E respectively, then the value of $12\left(e_H^2+e_E^2\right)$ is equal to ____. **[JEE (Main) 2022]**

MHB079

19. Let $a>0, b>0$. Let e and ℓ respectively be the eccentricity and length of the latus rectum of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. Let e' and ℓ' respectively the eccentricity and length of the latus rectum of its conjugate hyperbola. If $e^2=\frac{11}{14}\ell$ and $(e')^2=\frac{11}{8}\ell'$, then the value of $77a+44b$ is equal to **[JEE (Main) 2022]**

(1) 100 (2) 110 (3) 120 (4) 130

MHB080

20. Let the eccentricity of the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ be $\frac{5}{4}$. If the equation of the normal at the point $\left(\frac{8}{\sqrt{5}},\frac{12}{5}\right)$ on the hyperbola is $8\sqrt{5}x+\beta y=\lambda$, then $\lambda-\beta$ is equal to **[JEE (Main) 2022]**

MHB081

For More Join: @IITJEE_Advanced

21. Let $H_n = \frac{x^2}{1+n} - \frac{y^2}{3+n} = 1, n \in N$. Let k be the smallest even value of n such that the eccentricity of H_k is a rational number. If l is length of the latus return of H_k, then $21l$ is equal to.

[JEE (Main) 2023]

MHB082

22. The foci of a hyperbola are $(\pm 2,0)$ and its eccentricity is $\frac{3}{2}$. A tangent, perpendicular to the line $2x + 3y = 6$, is drawn at a point in the first quadrant on the hyperbola. If the intercepts made by the tangent on the x- and y-axes are a and b respectively, then $|6a| + |5b|$ is equal to.

[JEE (Main) 2023]

MHB083

23. Let $P(x_0, y_0)$ be the point on the hyperbola $3x^2 - 4y^2 = 36$, which is nearest to the line $3x + 2y = 1$. Then $\sqrt{2}\ (y_0 - x_0)$ is equal to: **[JEE (Main) 2023]**

(1) –3 (2) 9 (3) –9 (4) 3

MHB084

24. Let the eccentricity of an ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is reciprocal to that of the hyperbola $2x^2 - 2y^2 = 1$. If the ellipse intersects the hyperbola at right angles, then square of length of the latus-rectum of the ellipse is. **[JEE (Main) 2023]**

MHB085

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25. The vertices of a hyperbola H are $(\pm 6,0)$ and its eccentricity is $\frac{\sqrt{5}}{2}$. Let N be the normal to H at a point in the first quadrant and parallel to the line $\sqrt{2}x + y = 2\sqrt{2}$. If d is the length of the line segment of N between H and the y-axis then d^2 is equal to. **[JEE (Main) 2023]**

MHB086

26. Let $H : \frac{-x^2}{a^2} + \frac{y^2}{b^2} = 1$ be the hyperbola, whose eccentricity is $\sqrt{3}$ and the length of the latus rectum is $4\sqrt{3}$. Suppose the point $(\alpha, 6)$, $\alpha > 0$ lies on H. If β is the product of the focal distances of the point $(\alpha, 6)$, then $\alpha^2 + \beta$ is equal to: **[JEE (Main) 2024]**

(1) 170 (2) 171 (3) 169 (4) 172

MHB195

27. Consider a hyperbola H having centre at the origin and foci and the x-axis. Let C_1 be the circle touching the hyperbola H and having the centre at the origin. Let C_2 be the circle touching the hyperbola H at its vertex and having the centre at one of its foci. If areas (in sq. units) of C_1 and C_2 are 36π and 4π, respectively, then the length (in units) of latus rectum of H is **[JEE (Main) 2024]**

(1) $\frac{28}{3}$ (2) $\frac{14}{3}$ (3) $\frac{10}{3}$ (4) $\frac{11}{3}$

MHB196

For More Join: @IITJEE_Advanced

28. Let S be the focus of the hyperbola $\frac{x^2}{3}-\frac{y^2}{5}=1$, on the positive x-axis. Let C be the circle with its centre at $A\left(\sqrt{6},\sqrt{5}\right)$ and passing through the point S. if O is the origin and SAB is a diameter of C then the square of the area of the triangle OSB is equal to - **[JEE (Main) 2024]**

MHB197

29. Let P be a point on the hyperbola $H:\frac{x^2}{9}-\frac{y^2}{4}=1$, in the first quadrant such that the area of triangle formed by P and the two foci of H is $2\sqrt{13}$. Then, the square of the distance of P from the origin is **[JEE (Main) 2024]**

(1) 18 (2) 26 (3) 22 (4) 20

MHB198

30. If the foci of a hyperbola are same as that of the ellipse $\frac{x^2}{9}+\frac{y^2}{25}=1$ and the eccentricity of the hyperbola is $\frac{15}{8}$ times the eccentricity of the ellipse, then the smaller focal distance of the point $\left(\sqrt{2},\frac{14}{3}\sqrt{\frac{2}{5}}\right)$ on the hyperbola, is equal to **[JEE (Main) 2024]**

(1) $7\sqrt{\frac{2}{5}}-\frac{8}{3}$ (2) $14\sqrt{\frac{2}{5}}-\frac{4}{3}$ (3) $14\sqrt{\frac{2}{5}}-\frac{16}{3}$ (4) $7\sqrt{\frac{2}{5}}+\frac{8}{3}$

MHB199

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For More Join: @IITJEE_Advanced

EXERCISE - JEE (Advanced) PYQ

1. Let $P(6,3)$ be a point on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. If the normal at the point P intersects the x-axis at $(9,0)$, then the eccentricity of the hyperbola is - **[JEE (Advanced) 2011]**

(A) $\sqrt{\frac{5}{2}}$ (B) $\sqrt{\frac{3}{2}}$ (C) $\sqrt{2}$ (D) $\sqrt{3}$

MHB060

2. Tangents are drawn to the hyperbola $\frac{x^2}{9}-\frac{y^2}{4}=1$, parallel to the straight line $2x-y=1$. The points of contact of the tangents on the hyperbola are **[JEE (Advanced) 2012]**

(A) $\left(\frac{9}{2\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ (B) $\left(-\frac{9}{2\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$ (C) $(3\sqrt{3},-2\sqrt{2})$ (D) $(-3\sqrt{3},2\sqrt{2})$

MHB061

3. Consider the hyperbola $H: x^2-y^2=1$ and a circle S with centre $N(x_2,0)$. Suppose that H and S touch each other at a point $P(x_1,y_1)$ with $x_1>1$ and $y_1>0$. The common tangent to H and S at P intersects the x-axis at point M. If (l,m) is the centroid of the triangle ΔPMN, then the correct expression(s) is(are) **[JEE (Advanced) 2015]**

(A) $\frac{dl}{dx_1}=1-\frac{1}{3x_1^2}$ for $x_1>1$ (B) $\frac{dm}{dx_1}=\frac{x_1}{3\left(\sqrt{x_1^2-1}\right)}$ for $x_1>1$

(C) $\frac{dl}{dx_1}=1+\frac{1}{3x_1^2}$ for $x_1>1$ (D) $\frac{dm}{dy_1}=\frac{1}{3}$ for $y_1>0$

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MHB062

Triple Match : (Q. No. 4 to 6)

Column 1, 2 and 3 contain conics, equation of tangents to the conics and points of contact, respectively.

Column 1	Column 2	Column 3
(I) $x^2+y^2=a^2$	(i) $my = m^2x+a$	(P) $\left(\frac{a}{m^2},\frac{2a}{m}\right)$
(II) $x^2+a^2y^2=a^2$	(ii) $y=mx+a\sqrt{m^2+1}$	(Q) $\left(\frac{-ma}{\sqrt{m^2+1}},\frac{a}{\sqrt{m^2+1}}\right)$
(III) $y^2=4ax$	(iii) $y=mx+\sqrt{a^2m^2-1}$	(R) $\left(\frac{-a^2m}{\sqrt{a^2m^2+1}},\frac{1}{\sqrt{a^2m^2+1}}\right)$
(IV) $x^2-a^2y^2=a^2$	(iv) $y=mx+\sqrt{a^2m^2+1}$	(S) $\left(\frac{-a^2m}{\sqrt{a^2m^2-1}},\frac{-1}{\sqrt{a^2m^2-1}}\right)$

4. The tangent to a suitable conic (Column 1) at $\left(\sqrt{3},\frac{1}{2}\right)$ is found to be $\sqrt{3}x+2y=4$, then which of the following options is the only **CORRECT** combination? **[JEE (Advanced) 2017]**

(A) (II) (iii) (R) (B) (IV) (iv) (S) (C) (IV) (iii) (S) (D) (II) (iv) (R)

MHB063

For More Join: @IITJEE_Advanced

5. If a tangent to a suitable conic (Column 1) is found to be $y = x + 8$ and its point of contact is (8,16), then which of the following options is the only **CORRECT** combination? **[JEE (Advanced) 2017]**

(A) (III) (i) (P) (B) (III) (ii) (Q) (C) (II) (iv) (R) (D) (I) (ii) (Q)

MHB064

6. For $a=\sqrt{2}$, if a tangent is drawn to a suitable conic (Column 1) at the point of contact (– 1,1), then which of the following options is the only **CORRECT** combination for obtaining its equation?

[JEE (Advanced) 2017]

(A) (II) (ii) (Q) (B) (III) (i) (P) (C) (I) (i) (P) (D) (I) (ii) (Q)

MHB065

7. Let H: $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$, where $a > b > 0$, be a hyperbola in the xy-plane whose conjugate axis LM subtends an angle of 60° at one of its vertices N. Let the area of the triangle LMN be $4\sqrt{3}$.

LIST-I	**LIST-II**
P. The length of the conjugate axis of H is	1. 8
Q. The eccentricity of H is	2. $\frac{4}{\sqrt{3}}$
R. The distance between the foci of H is	3. $\frac{2}{\sqrt{3}}$
S. The length of the latus rectum of H is	4. 4

The correct option is : **[JEE (Advanced) 2018]**

(A) P → 4; Q → 2, R → 1; S → 3 (B) P → 4; Q → 3; R → 1; S → 2

(C) P → 4; Q → 1, R → 3; S → 2 (D) P → 3; Q → 4; R → 2; S → 1

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MHB066

8. Let a and b be positive real numbers such that $a > 1$ and $b < a$. Let P be a point in the first quadrant that lies on the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$. Suppose the tangent to the hyperbola at P passes through the point (1,0), and suppose the normal to the hyperbola at P cuts off equal intercepts on the coordinate axes. Let Δ denote the area of the triangle formed by the tangent at P, the normal at P and the x-axis. If e denotes the eccentricity of the hyperbola, then which of the following statements is/are TRUE? **[JEE (Advanced) 2020]**

(A) $1 < e < \sqrt{2}$ (B) $\sqrt{2} < e < 2$ (C) $\Delta = a^4$ (D) $\Delta = b^4$

MHB067

9. Consider the hyperbola $\frac{x^2}{100}-\frac{y^2}{64}=1$ with foci at S and S_1, where S lies on the positive x-axis. Let P be a point on the hyperbola, in the first quadrant. Let $\angle SPS_1 = \alpha$, with $\alpha < \frac{\pi}{2}$. The straight line passing through the point S and having the same slope as that of the tangent at P to the hyperbola, intersects the straight line S_1P at P_1. Let δ be the distance of P from the straight line SP_1, and $\beta = S_1P$. Then the greatest integer less than or equal to $\frac{\beta\delta}{9}\sin\frac{\alpha}{2}$ is_____. **[JEE (Advanced) 2022]**

MHB068

For More Join: @IITJEE_Advanced

ANSWER KEY

EXERCISE (O-1)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A	B	B	B	A	C	B	A	B	C
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B	A	B	D	B	A	A	C	B	B
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	B	A	C	D	B	C	C	B	D	A

EXERCISE (O-2)

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	A,B,C,D	C,D	B,C	C,D	A,B,C,D	A,C,D	B,C,D	B,C	A,B,C	B,C,D
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	B,D	A,B	A,B	A,B,C,D	A,B,C,D	A,C	A,B,C	A,B,C	B,C	B,D
Que.	21	22	23	24	25	26	27	28	29	
Ans.	B	A	D	A,B,C,D	A,B,C,D	A,D	0.25	2.00	C	
Que.	30									
Ans.	A→Q; B→S; C→S; D→R									

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EXERCISE (S)

1.	4	**2.**	2	**3.**	8	**4.**	5	**5.**	3
6.	9	**7.**	81.00	**8.**	45.00	**9.**	77	**10.**	$45\sqrt{5}$

EXERCISE - JEE (Main) PYQ

Que.	1	2	3	4	5	6	7	8	9	10
Ans.	2	3	4	4	3	3	2	2	2	1
Que.	11	12	13	14	15	16	17	18	19	20
Ans.	4	1	2	2	3	3	4	42	4	85
Que.	21	22	23	24	25	26	27	28	29	30
Ans.	306	12	3	2	216	2	1	40	3	1

EXERCISE - JEE (Advanced) PYQ

Que.	1	2	3	4	5	6	7	8	9	
Ans.	B	A,B	A,B,D	D	A	D	B	A,D	7	

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Important Notes

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